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Advanced cardiac life support (ACLS) in adults

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INTRODUCTION

Advanced cardiac life support (ACLS) guidelines have evolved over the past several decades based on a combination of scientific evidence of variable strength and expert consensus. The American Heart Association (AHA) and European Resuscitation Council (ERC) published the most recent ACLS Guidelines in 2025, using the comprehensive review of resuscitation literature performed by the International Liaison Committee on Resuscitation (ILCOR) [1,2]. Guidelines are reviewed continually, with formal updates published periodically in the journals Circulation and Resuscitation.

This topic will discuss the management of cardiac arrhythmias in adults as generally described in the most recent iteration of the ACLS Guidelines. Where our suggestions differ or expand upon the published guidelines, we state this explicitly. The evidence supporting the published guidelines is presented separately, as are issues related to basic life support, airway management, post-cardiac arrest management, pediatric resuscitation, and controversial treatments for cardiac arrest patients.

- **Basic resuscitation** (see "Adult basic life support (BLS) for health care providers" and "Basic airway management in adults")
- **Airway management** (see "Overview of advanced airway management in adults for emergency medicine and critical care" and "Extraglottic devices for emergency airway management in adults" and "Rapid sequence intubation in adults for emergency medicine and critical care" and "Emergency cricothyrotomy (cricothyroidotomy)")
- **Post-resuscitation care** (see "Initial assessment and management of the adult post-cardiac arrest patient" and "Intensive care unit management of the intubated post-cardiac arrest adult patient")
- **Resuscitation in specific settings** (see "Accidental hypothermia in adults: Clinical manifestations and evaluation" and "Drowning (submersion injuries)" and "Electrical injuries and lightning strikes: Evaluation and management" and "Initial management of the critically ill adult with an unknown overdose" and "Anaphylaxis: Emergency treatment")
- **Pediatric resuscitation** (see "Pediatric basic life support (BLS) for health care providers" and "Pediatric advanced life support (PALS)" and "Basic airway management in children")
- **Evidence and nonstandard treatments** (see "Supportive data for advanced cardiac life support in adults with sudden cardiac arrest" and "Therapies of uncertain benefit in basic and advanced cardiac life support")

EVIDENCE-BASED GUIDELINES

Because of the nature of resuscitation research, few randomized controlled trials have been completed in humans. Many of the recommendations in the Guidelines for ACLS and subsequent updates published jointly by the American Heart Association (AHA) and the International Liaison Committee on Resuscitation (ILCOR), hereafter referred to as the ACLS Guidelines, are made based upon observational studies, animal studies, and expert consensus [1,2]. Guideline recommendations are classified according to the GRADE system [3]. The evidence supporting the ACLS Guidelines is reviewed in detail separately. (See "Supportive data for advanced cardiac life support in adults with sudden cardiac arrest".)

PRINCIPLES OF MANAGEMENT

Excellent basic life support and its importance — Excellent cardiopulmonary resuscitation (CPR) and early defibrillation for shockable arrhythmias remain the cornerstones of basic life support and ACLS [4-9]. Although iterative updates for the ACLS Guidelines have suggested a number of revisions, including medications and monitoring, the emphasis on timely, excellent CPR and its critical role in resuscitative efforts remains unchanged (algorithm 1 and algorithm 2) [1,10]. The most recent versions of the ACLS algorithms can be accessed online here.

We emphasize the term "excellent CPR" because anything short of this standard does not achieve adequate cerebral and coronary perfusion, thereby compromising a patient's chances for neurologically intact survival. CPR is discussed in detail separately; key principles in the performance of ACLS are summarized in the following table (table 1). (See "Adult basic life support (BLS) for health care providers".)

Key principles in the performance of ACLS

Excellent CPR is crucial. • Anything short of excellent CPR does not achieve adequate cerebral and coronary perfusion. • Excellent chest compressions **must** be performed throughout the resuscitation without interruption, using proper timing (100 to 120 compressions per minute) and force (5 to 6 cm [2 to 2.5 inches] depth), and allowing for complete chest recoil. • Excellent chest compressions take priority over ventilation. • If a second rescuer is present, ventilations must be performed using proper timing (6 to 8 breaths per minute in the intubated patient; ratio of 30 compressions to 2 ventilations if not intubated) and force (deliver each breath over one second, and only until chest begins to rise). Avoid hyperventilation. • Do **not** stop compressions until the defibrillator is fully charged.
Defibrillate VF and pulseless VT as rapidly as possible.
Rapidly identify and treat causes of non-shockable arrest (PEA, asystole). • Important causes include the 5 H's and 5 T's: Hypoxia, Hypovolemia, Hydrogen ions (acidosis), Hyper/Hypo-kalemia, Hypothermia; Tension pneumothorax, Tamponade-cardiac, Toxins, Thrombosis-coronary (MI), Thrombosis-pulmonary (PE). • If immediately reversible causes (eg, tension pneumothorax, cardiac tamponade) are not corrected rapidly, the patient has little chance of survival.

Key principles in the performance of ACLS

Table 1 - larger image below

Studies in both the in-hospital and prehospital settings demonstrate that chest compressions are often performed incorrectly, inconsistently, and with excessive interruption [11-15]. To be effective, chest compressions must be of sufficient depth (5 to 6 centimeters, or 2 to 2.4 inches) and rate (between 100 and 120 per minute) and must allow for complete recoil of the chest between compressions.

Chest compression fraction, the proportion of total CPR time during which chest compressions are delivered, should be above 80 percent. In the past, clinicians frequently interrupted CPR to check for pulses, perform tracheal intubation, or obtain venous access. Current ACLS Guidelines strongly recommend that every effort be made **not** to interrupt CPR; interventions that have not been shown to improve outcomes, including tracheal intubation, venous access, and administration of medications to treat arrhythmias are carried out while CPR is performed. If the airway is obstructed, immediate management must be initiated and may necessitate interruption of compressions. (See "Airway foreign bodies in adults", section on 'Life-threatening asphyxiation' and "Emergency cricothyrotomy (cricothyroidotomy)".)

A single biphasic defibrillation shock remains the recommended treatment for ventricular fibrillation or pulseless ventricular tachycardia. CPR should be performed until the defibrillator is charged and resumed immediately after the shock is given, without pausing to recheck a pulse [16,17]. Assessment of waveform end-tidal carbon dioxide ($EtCO_2$) may be used as an adjunct to pulse checks if the patient is intubated

(receiving asynchronous ventilation); however, further study of its reliability is needed. Interruptions in CPR (eg, for subsequent attempts at defibrillation) should occur no more frequently than every two minutes and for the shortest possible duration. Compressions are paused briefly for ventilation when using a bag-valve-mask ventilation device at a ratio of 30:2. (See "Carbon dioxide monitoring (capnography)", section on 'Effectiveness of CPR'.)

There is a delay between the return of an organized electrical rhythm and effective myocardial contractions [18]. Thus, post-defibrillation pulse and rhythm checks are performed after two minutes of additional CPR. Key elements in the performance of manual defibrillation are described in the following table (table 2).

Patients are often overventilated during resuscitation, resulting in excessive intrathoracic pressure, which can compromise venous return and result in reduced cardiac output and inadequate cerebral and cardiac perfusion. Delivery of 30 compressions followed by two rescue breaths is recommended in patients without an advanced airway in place. ACLS Guidelines advise asynchronous ventilations 10 per minute if an endotracheal tube or extraglottic airway is in place, while continuous chest compressions are performed simultaneously [19].

Resuscitation team management — A growing body of literature demonstrates that employing the principles of Crisis Resource Management (CRM), adapted from the aviation industry and introduced into medical care by anesthesiologists, decreases disorganization during resuscitation and improves patient care [20-23]. A primary goal of CRM is to access the collective knowledge and experience of the team to provide the best care possible and to compensate for oversights or other challenges that any individual is likely to experience during such stressful events. Training in these principles to improve the quality of ACLS performed by health care clinicians is feasible and recommended [24,25].

Two principles provide the foundation for CRM: leadership and communication [22]. Resuscitations usually involve health care providers from different disciplines, sometimes from different areas of an institution, who may not have worked together previously. Under these circumstances, role clarity can be difficult to establish. In CRM, it is imperative that one person assumes the role of team leader [22]. This person is responsible for the global management of the resuscitation, including ensuring that all required tasks are carried out competently, assigning specific team members their responsibilities, incorporating new information and coordinating communication among all team members, developing and implementing management strategies that will maximize patient outcome, and reassessing performance throughout the resuscitation. Many clinical systems predetermine the leader for hospital resuscitation ("code") teams.

The team leader must avoid performing technical procedures, as performance of a task inevitably shifts attention from the primary leadership responsibilities. In circumstances where staff expertise is limited, the team leader may be required to perform certain critical procedures. In these situations, leadership is specifically transferred to another clinician, if possible, or the team leader may be forced temporarily to perform both roles, although this compromises the ability to provide proficient leadership and assimilate new information.

In CRM, communication is organized to provide effective and efficient care. All pertinent communication goes through the team leader, and the team leader shares important information with the team. When the team leader determines the need to perform a task, the request is directed to a specific team member, ideally by name. That team member verbally acknowledges the request and performs the task or, if unable to do so, informs the team leader that someone else should be assigned. Team members must be comfortable providing such feedback to the team leader. Specific emphasis is placed on the assigned team member repeating back medication doses and defibrillator energy settings to the team leader. This "closed-loop"

communication leads to a more orderly transfer of information and is the appropriate standard for all communication during resuscitations.

Though most decisions emanate from the team leader, a good team leader enlists the collective wisdom and experience of the entire team as needed. Team members must be encouraged to speak up if they have an observation, concern, or a feasible suggestion. Efforts should be made to overcome the tendency to withhold potentially lifesaving suggestions due to the fear of being incorrect or the nature of hierarchies that exist in many health care institutions. Extraneous personnel not directly involved with patient care are asked to leave to reduce noise and to ensure that orders from the leader and feedback from the resuscitation team can be heard clearly, and all noncritical verbalization must stop to ensure team harmony and clear communication.

INITIAL MANAGEMENT AND ECG INTERPRETATION

Beginning in the 2010 ACLS Guidelines, circulation assumed a more prominent role in the initial management of adult cardiac arrest, and this approach continues in subsequent iterations and updates. The "mantra" remains: circulation, airway, breathing (C-A-B). Once unresponsiveness is recognized, resuscitation begins by addressing circulation (excellent chest compressions), followed by airway opening, and then rescue breathing. In parallel, additional resources are mobilized by calling for help. Identifying a specific individual to call for help is more effective than a vague, general instruction for "someone" to do so. The ACLS Guidelines emphasize the importance of excellent, uninterrupted chest compressions and early defibrillation. Rescue breathing is performed after the initiation of excellent chest compressions. Advanced airway management may be delayed if there is adequate rescue breathing without an advanced airway in place. (See 'Excellent basic life support and its importance' above and "Adult basic life support (BLS) for health care providers", section on 'Recognition of cardiac arrest'.)

In the noncardiac arrest situation, the other initial interventions for ACLS include administering oxygen (if the patient's oxygen saturation is measurable and below 94 percent), establishing vascular access, placing the patient on a cardiac and oxygen saturation monitor, and obtaining an electrocardiogram (ECG) [26-28]. Unstable patients must receive immediate care, even when data are incomplete or presumptive (algorithm 1 and algorithm 2). The most recent versions of the ACLS algorithms can be accessed online here.

Patients with ST elevation myocardial infarction on ECG should be prepared for rapid transfer to the catheterization laboratory, receive a thrombolytic (if not contraindicated), or be transferred to a center with percutaneous coronary intervention capabilities. These decisions are made based on local resources and protocols.

Stable patients require an assessment of their ECG to provide appropriate treatment consistent with ACLS Guidelines. Although it is best to make a definitive interpretation of the ECG prior to making management decisions, the settings in which ACLS Guidelines are commonly employed require a modified, empirical approach. Such an approach is guided by the following questions:

- Is the rhythm fast or slow?
- Are the QRS complexes wide or narrow?
- Is the rhythm regular or irregular?

The answers to these questions often enable the clinician to make a provisional diagnosis and initiate appropriate therapy.

AIRWAY MANAGEMENT

In the minutes following sudden cardiac arrest, oxygen delivery is limited primarily by reduced blood flow, leading to the recommendation that excellent chest compressions take priority over ventilation during the initial resuscitation [4,5,29]. (See 'Principles of management' above.)

Suggested approach to airway management while performing ACLS — ACLS Guidelines support the use of a bag-valve-mask (BVM) device or placement of a supraglottic airway for ventilation during the initial management of sudden cardiac arrest unless one cannot ventilate the patient by these means or there is high certainty of rapid, successful placement of the tracheal tube without interruption of chest compressions [30-32]. Generally, endotracheal intubation can be deferred until after return of spontaneous circulation (ROSC). The performance of BVM ventilation is described in detail separately. (See "Basic airway management in adults".)

The ventilation rate is determined by whether the patient is intubated.

- If the patient is not intubated but ventilated using a BVM, the compression to ventilation ratio is 30:2. Although rescuers may be tempted to deliver non-synchronized BVM ventilations during cardiopulmonary resuscitation (CPR) to minimize interruptions in compressions, the mechanics of mask ventilations make it impossible to deliver adequate tidal volume during an active compression.

- If the patient is intubated, we suggest delivering 10 breaths per minute. Tidal volumes of no more than 600 mL delivered in a controlled fashion such that chest rise occurs over no more than one second is recommended in the ACLS Guidelines. (See "Adult basic life support (BLS) for health care providers", section on 'Ventilations'.)

  Overzealous ventilation (excess volume and/or frequency) elevates intrathoracic pressure, thereby decreasing venous return, ventricular filling, and stroke volume with compressions; all of which result in inadequate cerebral perfusion. In addition, overventilation can cause gastric inflation, which increases the risk of regurgitation and aspiration.

As a standard BVM for adults has a volume of 1000 to 1500 mL, even if some air is lost to the environment, a full squeeze of the bag during ventilation is unnecessary to deliver 600 mL.

Techniques and technical considerations — A blindly inserted extraglottic airway (eg, laryngeal mask airway, laryngeal tube, Combitube) can be placed without interrupting excellent chest compressions, provides adequate ventilation in most cases, and may reduce the risk of aspiration compared with BVM ventilation [33]. We believe that this is a reasonable approach, equal or superior to BVM ventilation. Extraglottic airways can be placed by basic providers and are considered alternatives to BVM ventilation, whereas tracheal intubation is an advanced technique for providers with the requisite training. Extraglottic airways and tracheal intubation are discussed separately. (See "Extraglottic devices for emergency airway management in adults", section on 'Extraglottic airway devices' and "Direct laryngoscopy and endotracheal intubation in adults".)

If intubation is to be performed during cardiac arrest, it must be done by a trained provider, ideally require less than 10 seconds to complete, be performed without interruption of chest compressions, and occur only after all other essential resuscitative maneuvers have been initiated. Multiple observational studies report better outcomes from video laryngoscopy compared with direct laryngoscopy in the setting of cardiac arrest, so video laryngoscopy is preferred when available. Once performed, rescuers must avoid hyperventilation. If ventilation is inadequate using a BVM or an extraglottic airway (eg, upper airway obstruction), intubation can be attempted during ongoing chest compressions or deferred to the two-minute interval (after a complete

cycle of CPR) when the resuscitator is already committed to stopping CPR for a rhythm check and possible defibrillation. If ventilation cannot be provided by BVM or an extraglottic airway because of apparent obstruction, the clinician must determine immediately whether arrest is due to upper airway obstruction and intervene as necessary. (See "Devices for difficult airway management in adults for emergency medicine and critical care".)

The ACLS Guidelines include the following additional recommendations about airway management during the performance of ACLS [34]:

- **Oxygen delivery** – It is reasonable to provide 100 percent oxygen during CPR. In patients with ROSC, oxygen concentration should be adjusted to maintain oxygen saturation (SpO_2) 90 percent. However, based on evidence presented below, we believe that oxygen should **not** be decreased rapidly in the early postarrest period to achieve this goal.

  Hyperoxia theoretically increases the risk of oxidative brain injury after ROSC, and significant early hyperoxia (eg, partial pressure of oxygen [PaO_2] >250mmHg) is associated with worse outcomes at hospital discharge in observational studies [35,36]. However, a randomized trial of 425 patients resuscitated from out-of-hospital cardiac arrest reported lower rates of survival to hospital discharge among those managed with rapid titration of inhaled oxygen to a goal SpO_2 of 90 to 94 percent compared with those managed with a goal SpO_2 of 98 to 100 percent (38 [82/214] versus 48 [101/211] percent, respectively; mean difference 10 percent, 95% CI 19-0.2) [37]. The low SpO_2 group also experienced twice the number of hypoxic episodes. (See "Initial assessment and management of the adult post-cardiac arrest patient", section on 'Mechanical ventilation' and "Overview of the acute management of ST-elevation myocardial infarction", section on 'Therapies of unclear benefit'.)

- **Cricoid pressure** should not be applied during intubation. It may be useful for preventing gastric insufflation during BVM ventilation. These issues are discussed separately. (See "Rapid sequence intubation in adults for emergency medicine and critical care", section on 'Positioning and protection'.)

- **Oropharyngeal and nasopharyngeal airways** – Both of these airways can improve the quality of BVM ventilation and should be used whenever possible. (See "Basic airway management in adults", section on 'Airway adjuncts'.)

- **Capnography** – Continuous waveform capnography (performed in addition to clinical assessment) is recommended for both confirming and monitoring correct tracheal tube placement and for monitoring the quality of CPR and ROSC. If waveform capnography is not available, a nonwaveform carbon dioxide (CO_2) detector may be used in addition to clinical assessment. (See "Carbon dioxide monitoring (capnography)", section on 'Clinical applications for intubated patients'.)

Evidence concerning airway management

Randomized trials – The optimal approach to airway management during sudden cardiac arrest remains uncertain, but it is likely BVM ventilation or an extraglottic airway, which are equally effective as tracheal intubation, more rapidly placed, and require less training [33,38,39].

- In a randomized trial of BVM ventilation (1020 patients) versus tracheal intubation (1023 patients) for prehospital management of out-of-hospital cardiac arrest in France or Belgium between 2015 and 2017, the primary outcome (survival with favorable neurologic outcome at 28 days) was similar in the two groups (4.3 percent for BVM compared with 4.2 percent for tracheal intubation) [40]. The trial failed to meet the prespecified criteria for noninferiority. Ambulance teams in these countries include physicians with training in intubation, which is not common in many countries.

- In a multicenter cluster randomized trial of a supraglottic airway device (4886 patients) versus tracheal intubation (4410 patients) for prehospital airway management of out-of-hospital cardiac arrest in England between 2015 and 2017, the primary outcome (favorable neurologic outcome at hospital discharge or 30 days, or at three- or six-month follow-up) was similar between the two groups [38,41]. There were no differences in survival at 72 hours or at 30 days. However, initial ventilation success occurred more commonly in the supraglottic airway group (87 versus 79 percent).

- In a multicenter cluster-crossover trial of a laryngeal tube (1505 patients) versus tracheal intubation (1499 patients) for prehospital airway management of out-of-hospital cardiac arrest in the United States between 2015 and 2017, the primary outcome (72-hour survival) occurred significantly more often in patients randomized to receive the laryngeal tube (18 versus 15 percent) [42]. Survival to discharge and functionally favorable survival were also greater in the laryngeal tube group.

- In a network meta-analysis of eight randomized and three quasi-randomized trials involving just under 16,000 patients, no difference in survival or neurologic outcome was found among the three approaches to prehospital airway management: supraglottic airway, BVM ventilation, and tracheal intubation [33]. Supraglottic airway placement was associated with a higher rate of ROSC. A meta-analysis of four randomized trials involving over 13,000 patients with out-of-hospital cardiac arrest reached similar conclusions [39].

Until additional data are available suggesting a clear improvement in patient-important outcomes from a particular ventilatory technique, BVM ventilation or placement of a supraglottic device (with close attention to avoiding overventilation) remains the preferred approach to airway management for cardiac arrest patients. (See 'Suggested approach to airway management while performing ACLS' above.)

Observational studies – The results of two large observational studies suggest that endotracheal intubation is not the best approach for managing patients with sudden cardiac arrest:

- In a prospective nationwide Japanese study involving 649,359 patients with sudden out-of-hospital cardiac arrest, the rate of survival with a favorable neurologic outcome was significantly lower among those managed with advanced airway techniques compared with BVM (1.1 versus 2.9 percent; odds ratio [OR] 0.38, 95% CI 0.36-0.39) [43]. Higher rates of survival with a favorable neurologic outcome when using BVM persisted across all analyzed subgroups, including adjustments for initial rhythm, ROSC, bystander CPR, and additional treatments.

- A study drawing on data collected between 2000 and 2014 from the Get With the Guidelines - Resuscitation multicenter registry used a propensity-matched cohort to compare outcomes among intubated and nonintubated patients who sustained in-hospital cardiac arrest [44]. In this study, each of the 43,314 patients intubated during the first 15 minutes of presentation following sudden cardiac arrest was matched with a patient not intubated in the same minute. Rates of ROSC (57.8 versus 59.3 percent), survival (16.3 versus 19.4 percent), and survival with good functional outcome (10.6 versus 13.6 percent) were all lower among intubated patients, and these findings held across all prespecified subgroup analyses.

Although both of these studies have limitations due to their observational nature and may not be generalizable to all settings, their size and consistent findings across all subgroup analyses support their conclusions.

When tracheal intubation is performed, multiple observational studies report better outcomes with video laryngoscopy compared with direct laryngoscopy:

- In a retrospective study of 22,132 patients treated by emergency medical service for out-of-hospital cardiac arrest, video laryngoscopy was independently associated with overall intubation success (adjusted OR 1.6, 95% CI 1.4-1.7) and first-pass success (adjusted OR 1.5, 95% CI 1.3-1.6) compared with direct laryngoscopy [45].

- A meta-analysis including the study above, five additional observational studies, and a small randomized trial found that video laryngoscopy was associated with improved first-pass success (OR 1.86, 95% CI 1.41-2.47), improved glottic view (OR 2.01, 95% CI 1.59-2.53), and fewer esophageal intubations (OR 0.25, 95% 0.08–0.85), compared with direct laryngoscopy, although no association with ROSC was observed [46].

MEDICATIONS USED DURING CPR

- **Epinephrine** – Epinephrine is the only medication indicated in sudden cardiac arrest regardless of arrest rhythm. Epinephrine is a sympathomimetic catecholamine that binds alpha-1, alpha-2, beta-1, and beta-2 receptors. During cardiopulmonary resuscitation (CPR), epinephrine is administered to increase systemic vasomotor tone via alpha-1 agonism, thereby increasing diastolic blood pressure and coronary perfusion pressure. The ACLS Guidelines recommend epinephrine (1 mg intravenous [IV] or intraosseous [IO] every three to five minutes) be administered after two minutes of CPR in shockable rhythms if initial rescue shocks are not successful.

  Some study results have raised doubts about the benefit of epinephrine [47-50]. In a randomized trial of 8014 patients who suffered out-of-hospital cardiac arrest, IV epinephrine increased the rate of return of spontaneous circulation (ROSC) compared with placebo (36 versus 12 percent) but did not lead to significant improvements in survival to hospital discharge with good neurologic outcome (2.2 versus 1.9 percent) [47]. This trial did not standardize or measure post-arrest care, potentially attenuating the benefit from improved ROSC in the epinephrine group. The results of this trial have not led to changes in the ACLS protocols from the American Heart Association or European Resuscitation Council.

- **Atropine** – Atropine is not recommended for the treatment of asystole or pulseless electrical activity [31,32]. For symptomatic bradycardia, the initial dose of atropine is 1 mg IV. This dose may be repeated every three to five minutes to a total dose of 3 mg. (See 'Approach to bradycardia' below.)

- **Amiodarone and lidocaine** — Evidence suggests that antiarrhythmic drugs provide little survival benefit in refractory ventricular tachycardia (VT) or ventricular fibrillation (VF) [5,9,29,31,32]. A randomized trial of 3026 patients with out-of-hospital VT/VF refractory to initial defibrillation compared IV or IO amiodarone, lidocaine, and placebo and found no differences in survival to hospital discharge or functionally favorable survival in the overall study population [51]. In patients with witnessed collapse, amiodarone or lidocaine resulted in improved survival compared with placebo (28 versus 28 versus 23 percent).

  The ACLS Guidelines state that antiarrhythmic drugs may be used in certain situations [31,32]. We suggest that antiarrhythmic drugs may be administered after a second unsuccessful defibrillation attempt in anticipation of a third shock, particularly among patients with witnessed arrest in whom time to administration may be shorter [52]. (See 'Refractory pulseless ventricular tachycardia or ventricular fibrillation' below.)

  When used, amiodarone (300 mg IV/IO bolus with a repeat dose of 150 mg IV as indicated) or lidocaine (1 to 1.5 mg/kg IV/IO bolus, then 0.5 to 0.75 mg/kg every 5 to 10 minutes) may be administered in VT/VF unresponsive to defibrillation, CPR, and epinephrine.

- **Magnesium** – Magnesium sulfate (2 g IV/IO bolus, may be repeated once if initial dose has no effect, followed by a maintenance infusion) is used to treat polymorphic VT consistent with torsade de pointes but is not recommended for routine use in adult cardiac arrest patients. (See 'Irregular wide complex' below and "Acquired long QT syndrome: Clinical manifestations, diagnosis, and management", section on 'Initial management'.)

- **Other medications**

  - **Vasopressin** – Outcomes of patients who receive vasopressin during CPR are not superior to those who receive epinephrine alone, so vasopressin administration is not recommended in the ACLS Guidelines [53]. Among patients who have suffered in-hospital cardiac arrest, three randomized controlled trials support administration of vasopressin (20 international units IV with each dose of epinephrine) together with glucocorticoids (methylprednisolone 40 mg IV once) as an adjunct to standard CPR [54-56]. Across trials, addition of vasopressin and glucocorticoid to standard care increased the rate of ROSC but did not consistently result in improved survival or functionally favorable recovery. Vasopressin and glucocorticoid administration are not recommended by ACLS Guidelines.

  - **Calcium** – Calcium chloride has both vasopressor and inotropic effects, but has not shown benefit when used to treat cardiac arrest [57-59]. A randomized trial of calcium chloride versus placebo during resuscitation of out-of-hospital cardiac arrest was terminated early because of a trend towards reduced rates of ROSC in patients receiving calcium [60]. Calcium chloride (1 g IV) should not be routinely administered during CPR but may be indicated in some special circumstances (eg, hyperkalemia, calcium channel blocker toxicity, hypermagnesemia). (See "Treatment and prevention of hyperkalemia in adults" and "Calcium channel blocker poisoning" and "Hypermagnesemia: Causes, clinical manifestations, evaluation, and treatment", section on 'Treatment'.)

  - **Sodium bicarbonate** – Sodium bicarbonate can mitigate acidosis and hyperkalemia that may incite or worsen during cardiac arrest. However, according to meta-analyses of randomized trials and observational studies involving a total of over 20,000 patients, routine sodium bicarbonate administration during CPR does not provide a benefit [61,62]. Selective use of sodium bicarbonate (50 to 100 mEq IV) may be reasonable when there is strong clinical suspicion or laboratory evidence of significant pre-existing metabolic acidosis or hyperkalemia. (See "Approach to the adult with metabolic acidosis", section on 'Overview of therapy' and "Bicarbonate therapy in lactic acidosis".)

MANAGEMENT OF SPECIFIC ARRHYTHMIAS

Immediate patient management is algorithmic and does not depend on cardiac rhythm, as detailed above. (See 'Initial management and ECG interpretation' above.)

- The potential sudden cardiac arrest victim is assessed for responsiveness, breathing, and presence of a pulse. For patients with effective respiration and a palpable pulse, treatment is determined by the ventricular rate (tachycardiac or bradycardia) and clinical assessment of overall stability. (See 'Arrhythmias with a pulse' below.)

- Pulseless patients are managed initially with cardiopulmonary resuscitation (CPR). Patients with pulseless ventricular tachycardia (VT) and ventricular fibrillation (VF) are defibrillated as rapidly as possible. Additional clinical considerations are discussed in greater detail below. (See 'Pulseless patient in sudden cardiac arrest' below.)

Pulseless patient in sudden cardiac arrest

Pulseless ventricular tachycardia and ventricular fibrillation — Pulseless VT and VF are nonperfusing rhythms emanating from the ventricles for which early identification is critical. Successful resuscitation of patients with VT/VF requires excellent CPR and rapid defibrillation. The American Heart Association (AHA) algorithm for the management of cardiac arrest can be accessed here (algorithm 2). The most recent versions of the ACLS algorithms can be accessed online here.

Excellent CPR is performed without interruption until the rescuer is ready to perform early defibrillation and is continued until return of spontaneous circulation (ROSC) is achieved. Treatable underlying causes should be identified and managed as quickly as possible (table 3) [34,63,64]. Agonal breathing or transient convulsive activity may accompany these dysrhythmias, and responders should not delay initiating CPR by misinterpreting these signs.

Begin performing excellent chest compressions as soon as cardiac arrest is recognized and continue while the defibrillator is being attached. If a defibrillator is not immediately available, continue CPR until one is obtained. As soon as a defibrillator is available, attach it to the patient (figure 1) and charge it while continuing CPR, then stop compressions to assess the rhythm and defibrillate if appropriate (eg, VT/VF is present). If asystole or pulseless electrical activity is present, continue CPR. If defibrillation is performed, resume CPR immediately and continue compressions until the next pulse and rhythm check two minutes later. (See "Supportive data for advanced cardiac life support in adults with sudden cardiac arrest", section on 'VF and pulseless VT'.)

Decreased time to defibrillation improves the likelihood of successful conversion to a perfusing rhythm and patient survival. For the monitored patient who sustains a witnessed VT/VF arrest, if a defibrillator is immediately available and defibrillator pads are in place, immediately charge the defibrillator and deliver a shock. The 10 seconds or fewer of CPR that might have been applied prior to the shock are unlikely to have generated any meaningful perfusion.

Biphasic defibrillators are recommended because of their increased efficacy at lower energy levels [65-67]. The ACLS Guidelines recommend that when employing a biphasic defibrillator, clinicians use the initial dose of energy recommended by the manufacturer (120 to 200 J). If this dose is not known, the maximal dose may be used. We suggest a first defibrillation at maximal energy for VT/VF. If a monophasic defibrillator is used, 360 J is the appropriate energy dose for initial and subsequent shocks.

ACLS Guidelines recommend the resumption of CPR immediately after defibrillation without checking for a pulse. This recommendation is made because effective cardiac contractility lags restoration of an organized electrical rhythm. Clinicians should stop compressions to perform a rhythm check only after two minutes of CPR, and not before the defibrillator is fully charged if the rhythm is VT/VF. (See "Adult basic life support (BLS) for health care providers", section on 'Phases of resuscitation' and "Adult basic life support (BLS) for health care providers", section on 'Defibrillation'.)

If VT/VF persists after at least one attempt at defibrillation and two minutes of CPR, administer epinephrine (1 mg intravenous [IV] or intraosseous [IO] every three to five minutes) while CPR is performed [30,68]. Premature treatment with epinephrine (within two minutes of defibrillation) has been associated with decreased survival [69]. VT/VF that persists after defibrillation may be treated with amiodarone or lidocaine. (See 'Medications used during CPR' above.)

Refractory pulseless ventricular tachycardia or ventricular fibrillation — Coronary artery disease and myocardial infarction are common causes of shock-refractory VT/VF. The likelihood of ROSC and favorable

recovery decreases over time as whole-body ischemia causes progressive end-organ damage. Few patients with CPR ongoing after 40 to 50 minutes will recover [70-73].

Defibrillation strategies — Defibrillation may be unsuccessful when insufficient energy transits the fibrillating ventricle. Modern biphasic defibrillators adapt to a range of patient characteristics that affect impedance to ensure adequate energy delivery. Nevertheless, if the vector of current between defibrillator pads does not fully capture the ventricles, VT/VF may persist. In such circumstances, changing the location of the defibrillator pads to the anteroposterior (AP) position from the anterior-lateral position (termed "vector change") or adding a second set of AP pads (termed "double sequential defibrillation") may improve the chances of successful defibrillation.

The 2025 AHA Guidelines conclude that the usefulness of vector change or double sequential defibrillation have not been established. In contrast, we feel the vector change strategy may be reasonable in patients with persistent VT/VF despite three consecutive defibrillation attempts.

In a trial of patients with VF/VT out-of-hospital cardiac arrest refractory to three consecutive defibrillation attempts with anterior and lateral pad placement, patients were randomly assigned to vector change, the addition of AP pads followed by double sequential defibrillation from both anterior-lateral and AP pad locations, or continued usual care [74]. The study was halted early because of low recruitment during the COVID-19 pandemic. The preliminary results were that both vector change and double sequential defibrillation improved the primary outcome of survival to hospital discharge compared with usual care (21.7 versus 30.4 versus 13.3 percent, respectively). Rates of VF termination and return of spontaneous circulation were also higher in both intervention arms.

Outside of a clinical trial, access to multiple defibrillators for a single patient may be limited, and their use adds complexity that might detract from high-quality CPR. In the absence of any proven benefit of double sequential defibrillation compared with vector change, and assuming a biphasic defibrillator is used, it is our opinion that vector change is preferable for the management of shock-refractory VF/VT in most situations.

Extracorporeal cardiopulmonary resuscitation — Patients with refractory VT/VF may achieve ROSC after coronary revascularization. Thus, there is substantial interest in use of venoarterial extracorporeal membrane oxygenation (VA-ECMO) initiated as an adjunct to conventional CPR [75]. VA-ECMO results in substantially better systemic perfusion and oxygen delivery than CPR and may be a useful bridge to coronary revascularization and myocardial recovery. VA-ECMO initiated during CPR is considered extracorporeal CPR (ECPR). The indications, implementation, and evidence for ECPR are reviewed separately:

- Implementation (see "Extracorporeal life support in adults: Management of venoarterial extracorporeal membrane oxygenation (VA ECMO)" and "Extracorporeal life support in adults: Management of venovenous extracorporeal membrane oxygenation (VV ECMO)")
- Evidence (see "Extracorporeal life support in adults: Management of venoarterial extracorporeal membrane oxygenation (VA ECMO)", section on 'Sudden cardiac arrest (extracorporeal cardiopulmonary resuscitation)')

Programs for effective delivery of ECPR are complex and resource-intensive, and they require expertise and substantial multidisciplinary coordination between prehospital and in-hospital providers [76]. Optimal patient selection and implementation strategies are uncertain. ECPR is most efficacious when initiated prior to development of severe global hypoxic-ischemic injury and as a bridge to intervention to reverse the inciting cause of arrest. Ideal patients have favorable arrest characteristics (eg, witnessed collapse,

immediate CPR, and short duration from collapse to cannulation), evidence of adequate intra-arrest perfusion (eg, end-tidal carbon dioxide [$EtCO_2$] greater than 10 mmHg, low presenting arterial lactate), and a presumed reversible etiology of arrest (eg, acute coronary syndrome, massive pulmonary embolism). ECPR is not addressed in detail in the 2025 ACLS guidelines. ACLS Guidelines for ECPR were last updated in 2023 and state ECPR may be considered for selected patients when feasible [77].

Asystole and pulseless electrical activity — Asystole is defined as a complete absence of electrical and mechanical cardiac activity. Pulseless electrical activity (PEA) is defined as any one of a heterogeneous group of organized ECG rhythms without sufficient mechanical contraction of the heart to produce a palpable pulse or measurable blood pressure. By definition, asystole and PEA are nonperfusing rhythms requiring immediate initiation of excellent CPR. These rhythms do not respond to defibrillation. The AHA algorithm for the management of cardiac arrest can be accessed here (algorithm 2). The most recent versions of the ACLS algorithms can be accessed online here.

In the ACLS Guidelines, asystole and PEA are addressed together because successful management for both depends on excellent CPR and rapid reversal of underlying causes, such as hypoxia, hyperkalemia, poisoning, and hemorrhage [34,63,64]. Epinephrine is administered as soon as is feasible after chest compressions are begun [4,5].

Asystole may result from a primary or secondary cardiac conduction abnormality, possibly due to end-stage tissue hypoxia and metabolic acidosis, or, rarely, from excessive vagal stimulation. It is crucial to identify and treat all potential secondary causes of asystole or PEA as rapidly as possible. As tension pneumothorax and cardiac tamponade make CPR ineffective and are often rapidly reversible, the clinician should not hesitate to perform immediate needle thoracostomy or pericardiocentesis if thought necessary. Delay in performing either procedure can worsen outcomes, and there is little chance either intervention will make the situation worse. The accompanying tables describe important secondary causes of cardiac arrest (table 3).

Treatable conditions associated with cardiac arrest

Condition	Common associated clinical settings
Acidosis	Diabetes, diarrhea, drug overdose, renal dysfunction, sepsis, shock
Anemia	Gastrointestinal bleeding, nutritional deficiencies, recent trauma
Cardiac tamponade	Post-cardiac surgery, malignancy, post-myocardial infarction, pericarditis, trauma
Hyperkalemia	Drug overdose, renal dysfunction, hemolysis, excessive potassium intake, rhabdomyolysis, major soft tissue injury, tumor lysis syndrome
Hypokalemia*	Alcohol abuse, diabetes mellitus, diuretics, drug overdose, profound gastrointestinal losses
Hypothermia	Alcohol intoxication, significant burns, drowning, drug overdose, elder patient, endocrine disease, environmental exposure, spinal cord disease, trauma
Hypovolemia	Significant burns, diabetes, gastrointestinal losses, hemorrhage, malignancy, sepsis, trauma
Hypoxia	Upper airway obstruction, hypoventilation (CNS dysfunction, neuromuscular disease), pulmonary disease
Myocardial infarction	Cardiac arrest
Poisoning	History of alcohol or drug abuse, altered mental status, classic toxidrome (eg, sympathomimetic), occupational exposure, psychiatric disease
Pulmonary embolism	Immobilized patient, recent surgical procedure (eg, orthopedic), peripartum, risk factors for thromboembolic disease, recent trauma, presentation consistent with acute pulmonary embolism
Tension pneumothorax	Central venous catheter, mechanical ventilation, pulmonary disease (eg, asthma, chronic obstructive pulmonary disease), thoracentesis, thoracic trauma

Treatable conditions associated with cardiac arrest

Table 3 - larger image below

After initiating CPR, immediately consider and treat reversible causes as appropriate and administer epinephrine (1 mg IV every three to five minutes) as soon as feasible [5,30,68]. As with VT/VF, studies of epinephrine in patients with asystole or PEA report mixed results, and further study is needed [30,47,78]. Neither asystole nor PEA responds to defibrillation. Atropine is no longer recommended for the treatment of asystole or PEA. Cardiac pacing is ineffective for cardiac arrest and not recommended. Evidence around the management of asystole and PEA, and cardiac arrest generally, is reviewed in detail separately. (See "Supportive data for advanced cardiac life support in adults with sudden cardiac arrest".)

Intra-arrest monitoring — ACLS Guidelines encourage the use of clinical and physiologic monitoring to optimize performance of CPR and to detect ROSC [26]. Assessment and immediate feedback about the rate and depth of chest compressions, adequacy of chest recoil between compressions, and rate and force of ventilations improve CPR. These parameters should be monitored continuously and any necessary adjustments made immediately. Accelerometers have been integrated into several brands of defibrillator pads or freestanding devices that can be placed on the patient's sternum during chest compressions to provide these metrics and real-time feedback.

$EtCO_2$ measured from continuous waveform capnography can provide a rough estimate of cardiac output (and therefore the quality of CPR). $EtCO_2$ less than 10mmHg suggests inadequate cardiac output and the need to improve CPR quality or provide other interventions such as needle thoracostomy. Sudden, sustained increases in $EtCO_2$ >10 mmHg during CPR likely indicate ROSC. (See "Carbon dioxide monitoring (capnography)", section on 'Effectiveness of CPR' and "Carbon dioxide monitoring (capnography)", section on 'Return of spontaneous circulation'.)

Data from other physiologic monitors are less likely to be available in patients with sudden cardiac arrest, but measurements obtained from arterial catheters already in place can provide useful feedback about the quality of CPR and ROSC [34]. CPR should not be interrupted to place arterial or central venous catheters. Arterial diastolic pressure is a reasonable proxy for coronary perfusion pressure. A reasonable goal is to maintain an arterial diastolic pressure above 20 mmHg.

In the hands of skilled operators, point-of-care ultrasound may be useful during cardiac arrest for identifying underlying pathology, monitoring resuscitation, and determining the presence of cardiac activity and likelihood of recovery [79,80]. However, studies of point-of-care ultrasound in the setting of cardiac arrest are preliminary, and high-quality trials are needed. While such research is ongoing, it is crucial that ultrasound-related interventions not cause interruptions or otherwise interfere with the performance of excellent CPR.

Arrhythmias with a pulse

Bradycardia

Definition and clinical findings — Bradycardia is defined as a heart rate below 60 beats per minute, but symptomatic bradycardia generally entails rates below 40 beats per minute. The ACLS Guidelines recommend that clinicians not intervene unless the patient exhibits evidence of inadequate tissue perfusion thought to result from the slow heart rate [34,63,64]. Signs and symptoms of inadequate perfusion include hypotension, lightheadedness or a presyncopal sensation, altered mental status (including syncope), signs of shock, ongoing ischemic chest pain, and evidence of acute pulmonary edema. Hypoxia is a common cause of bradycardia. If peripheral perfusion is adequate, pulse oximetry should be used to assess oxyhemoglobin saturation. If perfusion is inadequate or pulse oximetry is unavailable, assess the patient for signs of respiratory failure (eg, increased or decreased respiratory rate, diminished respiratory volume, retractions, or paradoxical abdominal breathing). Bradycardia in the intubated patient should be considered to represent a malpositioned or displaced endotracheal tube until proven otherwise.

Approach to bradycardia — The AHA algorithm for the management of bradycardia can be accessed here (algorithm 3). The most recent versions of the ACLS algorithms can be accessed online here.

We generally administer atropine while simultaneously preparing for prompt temporary cardiac pacing (transvenous, if immediately available, or transcutaneous) and/or infusion of a chronotropic agent for bradycardic patients with clinically significant symptoms thought to be due to one of the following etiologies:

- High vagal tone (eg, inferior myocardial ischemia due to acute coronary syndrome)
- Medication-induced (supratherapeutic levels of beta blockers, calcium channel blockers, digitalis)
- High-degree atrioventricular (AV) block with a narrow QRS complex (thought to emanate at or above the AV node)

If bradycardia is thought to be due to a conduction disturbance at or below the bundle of His (characterized by a wide QRS complex in complete heart block or Mobitz type II second-degree AV block), we avoid atropine and proceed directly to cardiac pacing and/or administration of a chronotropic agent.

- **Atropine** – The initial dose of atropine is 1 mg IV. This dose may be repeated every three to five minutes to a total dose of 3 mg. (See "Second-degree atrioventricular block: Mobitz type II" and "Acquired third-degree (complete) atrioventricular block".)

- **Temporary pacing** – If temporary transvenous cardiac pacing can be initiated promptly, prepare for transvenous pacing, and obtain appropriate consultation as available. If transvenous pacing cannot be initiated promptly, initiate transcutaneous pacing, and prepare for chronotropic infusion. Before using transcutaneous pacing, assess whether the patient can perceive the pain associated with this procedure, and if so, provide appropriate sedation and analgesia whenever possible. (See "Procedural sedation in adults in the emergency department: General considerations, preparation, monitoring, and mitigating complications".)

  Patients requiring transcutaneous or transvenous pacing generally require cardiology consultation and admission for evaluation for possible permanent pacemaker placement unless a reversible cause of bradycardia, such as hyperkalemia or overmedication with a beta blocker or calcium channel blocker is identified and corrected.

- **Chronotropic agents** – For patients who remain symptomatic following atropine administration and for whom temporary cardiac pacing is either not readily available or not successful in alleviating symptoms, continuous infusion of a chronotropic agent is indicated. Either dopamine or epinephrine, but not both, should be initiated. Because of its superior vasoconstrictive effects, we prefer epinephrine as a first-line chronotropic agent when there is concomitant hypotension. The starting dose for infusions of dopamine is from 5 to 20 mcg/kg per minute, while epinephrine is started at 0.025 to 0.125 mcg/kg per minute (2 to 10 mcg per minute). Each should be titrated to the patient's response.

Tachycardia — Tachycardia is defined as a heart rate above 100 beats per minute, but symptomatic tachycardia generally involves rates over 150 beats per minute unless underlying ventricular dysfunction exists [34,63,64]. Management of tachyarrhythmias is governed by the presence of clinical symptoms and signs caused by the rapid heart rate. The AHA algorithm for the management of tachycardia can be accessed here (algorithm 4). The most recent versions of the ACLS algorithms can be accessed online here.

Approach to tachycardia — The fundamental approach is as follows: First, determine if the patient is unstable (eg, manifests ongoing ischemic chest pain, acute mental status changes, hypotension, signs of shock, or evidence of acute pulmonary edema). Hypoxemia is a common cause of unstable tachycardia; look for signs of labored breathing (eg, increased respiratory rate, retractions, paradoxical abdominal breathing) or low oxygen saturation.

If instability is present and appears related to the tachycardia, treat immediately with synchronized cardioversion unless the rhythm is sinus tachycardia [81]. Some cases of supraventricular tachycardia (SVT) may respond to immediate treatment with a bolus of adenosine (6 or 12 mg IV) without the need of cardioversion. Whenever possible, assess whether the patient can perceive the pain associated with cardioversion, and if so, provide appropriate sedation and analgesia if time permits. (See "Procedural sedation in adults in the emergency department: General considerations, preparation, monitoring, and mitigating complications".)

In the stable patient, use the ECG to determine the nature of the arrhythmia. In the urgent settings where ACLS algorithms are most often employed, specific rhythm identification may not be possible. Nevertheless, by performing a systematic review of the ECG, one can determine appropriate management. Three questions provide the basis for assessing the ECG in this setting:

- Is the patient in a sinus rhythm?
- Is the QRS complex wide or narrow?
- Is the rhythm regular or irregular?

More detailed approaches to rhythm determination in tachycardia are discussed separately. (See "Narrow QRS complex tachycardias: Clinical manifestations and evaluation of the electrocardiogram" and "Wide QRS complex tachycardias: Approach to the diagnosis" and "Overview of the acute management of tachyarrhythmias in adults".)

Regular narrow complex — A narrow QRS complex implies that a tachycardic rhythm originates at or above the AV node. SVT, including sinus tachycardia, is the major cause of a regular narrow complex arrhythmia [34,63,64]. Sinus tachycardia is a common response to fever, anemia, shock, sepsis, pain, heart failure, or any other physiologic stress. No medication is needed to treat sinus tachycardia; care is focused on treating the underlying cause. (See "Sinus tachycardia: Evaluation and management".)

Reentrant SVT is a regular tachycardia most often caused by a reentrant mechanism within the conduction system (algorithm 4). The QRS interval is usually narrow but can be longer than 120 ms if a bundle branch block (ie, SVT with rate-related aberrancy or fixed bundle branch block) is present. Vagal maneuvers slow conduction through the AV node and may interrupt the reentrant circuit. They may be employed on appropriate patients while other therapies are prepared. Vagal maneuvers alone (eg, Valsalva, carotid sinus massage) convert up to 25 percent of SVTs to sinus rhythm, while Valsalva followed immediately by supine repositioning with a passive leg raise has been shown to be even more effective. SVT refractory to vagal maneuvers is treated with adenosine [82-84]. (See "Overview of the acute management of tachyarrhythmias in adults" and "Narrow QRS complex tachycardias: Clinical manifestations and evaluation of the electrocardiogram" and "Reentry and the development of cardiac arrhythmias" and "Vagal maneuvers".)

Because of its extremely short half-life, adenosine (6 or 12 mg IV) is injected as rapidly as possible into a proximal vein followed immediately by a 20 mL saline flush and elevation of the extremity to ensure the drug enters the central circulation before it is metabolized. If the first dose of adenosine does not convert the rhythm, a second and third dose of 12 mg IV may be given. Larger doses (eg, 18 mg IV) may be needed in patients taking theophylline or theobromine or those who consume large amounts of caffeine; smaller doses (eg, 3 mg IV) should be given to patients taking dipyridamole or carbamazepine and those with transplanted hearts, or when injecting via a central vein.

Prior to injection, warn the patient about transient side effects from adenosine, including dysphoria, chest discomfort, dyspnea, and flushing, and give reassurance that these effects are very brief. Perform continuous ECG recording during administration. If adenosine fails to convert the SVT, consider other etiologies for this rhythm, including atrial flutter or a non-reentrant SVT, which may become apparent on the ECG when AV nodal conduction is slowed.

If conversion attempts fail and the patient remains stable, initiate rate control with either an IV nondihydropyridine calcium channel blocker or a beta blocker. Agents to choose from include diltiazem, verapamil, and any of several beta blockers (including metoprolol, atenolol, esmolol, and labetalol). (See "Atrial fibrillation in adults: Rate control in patients without heart failure", section on 'Urgent therapy'.)

Irregular narrow complex — Irregular narrow-complex tachycardias may be caused by atrial fibrillation, atrial flutter with variable AV nodal conduction, multifocal atrial tachycardia (MAT), or sinus tachycardia with frequent premature atrial complexes (PACs; also referred to as premature atrial beats, premature supraventricular complexes, or premature supraventricular beats) (algorithm 4). Of these, atrial fibrillation is most common [34,63,64].

The initial goal of treatment in stable patients is to control the heart rate using either a nondihydropyridine calcium channel blocker (diltiazem 10 to 20 mg IV over two minutes, repeat at 20 to 25 mg IV after 15 minutes; or verapamil 2.5 to 5 mg IV over two minutes followed by 5 to 10 mg IV every 15 to 30 minutes) or a beta blocker (eg, metoprolol 5 mg IV for three doses every two to five minutes, then up to 200 mg by mouth every 12 hours). The management of atrial fibrillation and SVT is discussed in detail separately. (See "Atrial fibrillation: Overview of management" and "Management of atrial fibrillation: Rhythm control versus rate control" and "Atrial fibrillation in adults: Rate control in patients without heart failure" and "Narrow QRS complex tachycardias: Clinical manifestations and evaluation of the electrocardiogram" and "Multifocal atrial tachycardia".)

Calcium channel blockers and beta blockers may cause or worsen hypotension. Patients should be closely monitored while the drug is given, and patients at greater risk of developing severe hypotension (eg, older adults) may require loading doses that are below the usual range. Adequate IV access should be established in case hypotension develops. Combination therapy with a beta blocker and calcium channel blocker increases the risk of severe heart block.

Diltiazem is suggested in most instances for the management of acute atrial fibrillation with rapid ventricular response. Beta blockers may also be used and may be preferred in the setting of an acute coronary syndrome. Beta blockers are more effective for chronic rate control. For atrial fibrillation associated with hypotension, amiodarone may be used (150 mg IV over 10 minutes followed by 1 mg/min drip for six hours, and then 0.5 mg/min) but may cause conversion to sinus rhythm, which may result in embolic injury if the atrial fibrillation was not short lived [85]. For atrial fibrillation associated with acute heart failure, amiodarone or digoxin may be used for rate control. Treatment of MAT includes correction of possible precipitants, such as hypokalemia and hypomagnesemia. The ACLS Guidelines recommend consultation with a cardiologist for these arrhythmias.

Cardioversion of stable patients with irregular narrow complex tachycardias should **not** be undertaken without considering the risk of embolic stroke. If the duration of atrial fibrillation is known to be less than 48 hours or the patient has been receiving long-term therapeutic anticoagulation (eg, warfarin with an international normalized ratio known to be therapeutic or a direct oral anticoagulant with good adherence), the risk of embolic stroke is low, and the clinician may consider electrical or chemical cardioversion [86]. Several medications may be used for chemical cardioversion, and the best drug varies according to clinical circumstances. The questions of whether chemical cardioversion is appropriate and which agent to select are reviewed separately.

Regular wide complex — A regular wide-complex tachycardia is generally ventricular in etiology (algorithm 4). Aberrantly conducted SVTs may also be seen. Because differentiation between VT and SVT with aberrancy can be difficult, assume VT is present. Treat clinically stable undifferentiated wide-complex tachycardia with antiarrhythmics or planned synchronized cardioversion [34,63,64].

In cases of regular wide-complex tachycardia with a monomorphic QRS complex, adenosine may be used for diagnosis and treatment. Do **not** give adenosine (or other AV nodal blocking medications) to patients who are unstable or manifest wide-complex tachycardia with an irregular rhythm or a polymorphic QRS

complex. SVT with aberrancy, if **definitively** identified (eg, old ECG demonstrates bundle branch block), may be treated in the same manner as narrow-complex SVT, with vagal maneuvers, adenosine, or rate control (see 'Irregular narrow complex' above). Adenosine is likely to slow or convert SVT with aberrancy. Dosing is identical to that used for SVT. Adenosine also terminates some cases of VT, particularly those that originate in the left or right ventricular outflow tracts [87]. Thus, adenosine responsiveness cannot be used to confirm a diagnosis of SVT or to exclude VT. (See 'Regular narrow complex' above.)

Other antiarrhythmic drugs that may be used to treat stable patients with regular wide-complex tachycardia include procainamide (20 to 50 mg/min IV) and amiodarone (150 mg IV given over 10 minutes, repeated as needed to a total of 2.2 g IV over the first 24 hours). A procainamide infusion continues until the arrhythmia is suppressed, the patient becomes hypotensive, the QRS widens 50 percent beyond baseline, or a maximum dose of 17 mg/kg is administered. Procainamide should be avoided in patients with a prolonged QT interval. If the wide-complex tachycardia persists despite pharmacologic therapy, cardioversion may be needed. The ACLS Guidelines recommend expert consultation for all patients with wide complex tachycardia.

Irregular wide complex — A wide-complex, irregular tachycardia may be atrial fibrillation with pre-excitation (eg, Wolf Parkinson White syndrome), atrial fibrillation with aberrancy (bundle branch block), or polymorphic VT/torsades de pointes (algorithm 4) [34,63,64]. Use of AV nodal blockers in wide-complex, irregular tachycardia of unclear etiology may precipitate VF and death and is contraindicated. Such medications include beta blockers, calcium channel blockers, digoxin, and adenosine. To avoid inappropriate and possibly dangerous treatment, the ACLS Guidelines suggest assuming that any wide-complex, irregular tachycardia is caused by pre-excited atrial fibrillation.

Adult tachycardia with a pulse algorithm 2020 update

Algorithm 4 - larger image below

Patients with a wide-complex, irregular tachycardia caused by pre-excited atrial fibrillation usually manifest extremely fast heart rates (generally over 200 beats per minute) and require immediate synchronized electric cardioversion. In cases where electric cardioversion is ineffective or unfeasible, or atrial fibrillation recurs, antiarrhythmic therapy with procainamide may be given. The ACLS Guidelines recommend expert consultation for all patients with wide-complex tachycardia. Dosing for antiarrhythmic medications is described above. (See 'Regular wide complex' above.)

Treat polymorphic VT with emergency defibrillation. Interventions to prevent recurrent polymorphic VT include correcting underlying electrolyte abnormalities (eg, hypokalemia, hypomagnesemia) and, if a prolonged QT interval is observed or thought to exist, stopping all medications that increase the QT interval. Magnesium sulfate (2 g IV administered via rapid IV bolus, may be repeated once if initial dose has no effect, followed by a maintenance infusion) can be given to prevent polymorphic VT associated with familial or acquired prolonged QT syndrome [88].

A clinically stable patient with atrial fibrillation and a wide QRS interval **known** to stem from a pre-existing bundle branch block (ie, old ECG demonstrates pre-existing block with the same QRS morphology) may be treated in the same manner as a narrow-complex atrial fibrillation.

Vascular access for medication administration — Based on evidence presented below, intravenous (IV) vascular access is the preferred first-line approach for ACLS medication administration, and intraosseous (IO) access is a reasonable alternative when IV is not feasible or is delayed. Medication doses for IO administration are identical to those for IV, although the pharmacokinetics of IV administration during cardiac arrest are more favorable. Tracheal administration of medications is not recommended. (See "Intraosseous infusion" and 'Alternative methods for medication administration' below.)

Vascular access via the IO route is safe and more easily initiated in the setting of cardiac arrest [34,63,64,89]. However, evidence from observational studies suggests that administration of medication via the IV route produces more favorable outcomes [90-93]. Nevertheless, when IV access cannot be established or its theoretical benefit is mitigated by the time and resources necessary to initiate it, IO lines are a safe and effective alternative.

Several randomized trials have compared a strategy of first-line attempt at IV versus IO access during out-of-hospital cardiac arrest:

- A systematic review and meta-analysis identified three randomized trials that enrolled a total of 9332 participants with emergency medical services (EMS)-treated out-of-hospital cardiac arrest (OHCA) in the United Kingdom, Taiwan, and Denmark, each of which compared a strategy of IO-first to IV-first vascular access [94]. In each study, successful intervention placement was more common for IO-first participants (92 to 94 percent) than for IV-first participants (58 to 80 percent), but the time to first drug administration did not vary by treatment arm. For the primary meta-analysis outcome of 30-day survival, there was no difference between treatment arms (odds ratio 0.99, 95% CI 0.84 to 1.17). However, in two studies (7545 participants) that reported sustained return of spontaneous circulation, this outcome was less frequent in patients randomized to receive IO-first (23 versus 25 percent).
- The largest trial, PARAMEDIC-3, enrolled 6082 patients with EMS-treated OHCA in the United Kingdom and randomly assigned patients to an IO-first or IV-first treatment strategy [95]. No statistically significant difference was found in survival to 30 days (4.5 percent IO-first group versus 5.1 percent IV-first group). However, more frequent return of spontaneous circulation was noted in the IV group (36 versus 39.1 percent).

Alternative methods for medication administration — Historically, tracheal administration of medications was recommended in patients without vascular access. This approach is now considered to be of questionable safety and efficacy, and it is no longer recommended.

CONSCIOUS PATIENT BEING RESUSCITATED

A minority of patients may be conscious while being resuscitated for cardiac arrest [31]. Cardiopulmonary resuscitation-induced consciousness (CPRIC) is a term used to describe this condition. The incidence, management, and consequences of CPRIC remain unclear and controversial. The safety and effectiveness of interventions (eg, sedation) intended to address this condition during resuscitation are unstudied. Clinicians seeking to intervene must weigh the theoretical benefits against the potential risks. Theoretical benefits include patient comfort, reducing the risk of invasive device dislodgement, and improving long-term psychologic outcomes, while potential risks include altered hemodynamics or respiratory depression, which may affect the likelihood of return of spontaneous circulation, rearrest, or postarrest hypotension or hypoxia.

USE OF ULTRASOUND AND ECHOCARDIOGRAPHY

Bedside echocardiography must never interfere with resuscitation efforts and should not interrupt or delay resumption of cardiopulmonary resuscitation (CPR) except in cases where the ultrasound is being obtained strictly to confirm absence of cardiac activity when a decision to terminate resuscitative efforts is imminent. The 2020 update of the ACLS Guidelines suggests that point-of-care ultrasound and echocardiography be employed to help identify reversible causes of cardiac arrest (eg, cardiac tamponade, tension pneumothorax, pulmonary embolism) and to assist in the identification of return of spontaneous circulation (ROSC) [4,96,97]. (See 'Termination of resuscitative efforts' below.)

According to a systematic review of 12 small trials, most of which studied convenience samples of patients with sudden cardiac arrest (n = 568), bedside echocardiography may be helpful to predict ROSC [98]. In this review, the pooled sensitivity and specificity of echocardiography to predict ROSC were 91.6 and 80 percent, respectively (95% CI for sensitivity 84.6-96.1 percent; 95% CI for specificity 76.1-83.6 percent). Of the 190 patients found to have cardiac activity, 98 (51.6 percent) achieved ROSC, whereas only 9 (2.4 percent) of the 378 with cardiac standstill did so. Other studies have reached similar conclusions about the rarity of ROSC in cases with cardiac standstill on ultrasound [99-101]. Echocardiography should **not** be the sole basis for terminating resuscitative efforts but may serve as an adjunct to clinical assessment.

RESUSCITATION OF PATIENTS WITH COVID-19 OR SIMILAR ILLNESS

Guidance for the performance of cardiopulmonary resuscitation (CPR) in patients with suspected or confirmed coronavirus disease 2019 (COVID-19)-related illness was first published by the American Heart Association (AHA) in 2020 and updated in 2021 [102,103]. Original and updated guidance emphasizes several key points, many of which apply also to other highly infectious and dangerous respiratory diseases:

- Vaccination against severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) offers significant protection to health care providers, including those involved in resuscitation of patients with suspected or confirmed COVID-19.
- Don personal protective equipment according to local guidelines and availability.

Systems and procedures should be in place to minimize any time delays in providing lifesaving interventions. Tasks and modifications for clinicians emphasized in the COVID-19-related guidelines include the following:

- Minimize the number of providers performing resuscitation.
- In the hospital setting, use a negative-pressure room whenever possible; keep the door to the resuscitation room closed if possible.
- A mechanical device may be used to perform chest compressions on adults and adolescents who meet minimum height and weight requirements.
- Use a high-efficiency particulate air filter for bag-mask ventilation and mechanical ventilation.

POST-RESUSCITATION CARE

The ACLS Guidelines recommend a combination of goal-oriented interventions provided by an experienced multidisciplinary team for all cardiac arrest patients with return of spontaneous circulation [4,7,63,104]. Important objectives for such care include:

- Optimizing cardiopulmonary function and perfusion of vital organs

- Managing acute coronary syndromes
- Implementing strategies to prevent and manage organ system dysfunction and brain injury

Management of the post-cardiac arrest patient is reviewed separately. (See "Initial assessment and management of the adult post-cardiac arrest patient" and "Intensive care unit management of the intubated post-cardiac arrest adult patient".)

TERMINATION OF RESUSCITATIVE EFFORTS

Criteria for determining whether to stop — On balance, termination of resuscitation (TOR) is ethical when the likelihood of achieving a favorable outcome with continued efforts is extremely low [105]. Determining when to stop resuscitation efforts in cardiac arrest patients is difficult, and little high-quality evidence exists to guide decision-making [106]. Furthermore, decision-making may vary depending on clinical circumstances, including settings discussed in the following topics. (See "Drowning (submersion injuries)" and "Accidental hypothermia in adults: Clinical manifestations and evaluation" and "Electrical injuries and lightning strikes: Evaluation and management" and "Initial management of the critically ill adult with an unknown overdose".)

Physician survey data and clinical practice guidelines suggest that factors influencing the decision to stop resuscitative efforts include [107-111]:

- Duration of resuscitative effort >30 minutes without a sustained perfusing rhythm
- Unwitnessed collapse with an initial ECG rhythm of asystole
- Prolonged interval between time of collapse and initiation of cardiopulmonary resuscitation (CPR)
- Patient age, severe comorbid disease, or prior functional dependence

The 2025 American Heart Association (AHA) ACLS guidelines provide separate guidance about TOR for basic life support (BLS) and advanced life support (ALS) providers. These incorporate evidence from multiple studies that have established that survival among patients meeting TOR rules is extremely uncommon [112].

TOR at the scene of arrest during BLS care is considered reasonable when **all** of the following criteria are met and resuscitation has been attempted for 20 to 30 minutes:

- Cardiac arrest was not witnessed by emergency medical services personnel
- There is no return of spontaneous circulation
- No defibrillations have been delivered

The AHA guidelines do not specify a minimum duration of resuscitation necessary to apply these rules, and premature application (eg, after only six minutes of attempted resuscitation) may result in TOR for patients who would otherwise survive [113]. By contrast, survival when these rules are applied after 20 to 30 minutes of resuscitation is exceedingly low [113,114].

TOR at the scene of arrest during ALS care is considered reasonable when **all** of the following criteria are met and resuscitation has been attempted for 20 to 30 minutes:

- The arrest was not witnessed
- The was no layperson CPR attempted
- There is no return of spontaneous circulation
- No defibrillations have been delivered

Again, the AHA guidelines do not specify a minimum duration of attempted resuscitation prior to applying the ALS TOR criteria. We believe 20 to 30 minutes of on-scene ACLS care is reasonable prior to TOR.

Discussion with family members — Guidance for breaking bad news or holding difficult discussions with the patient's family is provided separately. (See "Palliative care for adults in the ED: Goals of care, communication, consultation, and patient death", section on 'Communicating difficult news'.)

SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See "Society guideline links: Basic and advanced cardiac life support in adults".)

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topic (see "Patient education: Ventricular fibrillation (The Basics)")

SUMMARY AND RECOMMENDATIONS

- **Key principles and access to algorithms** – High-quality chest compressions and early defibrillation for treatable arrhythmias remain the cornerstones of basic life support and advanced cardiac life support (ACLS). The most recent versions of the ACLS algorithms can be accessed online; access to copies within UpToDate is provided below. (See 'Excellent basic life support and its importance' above.)
  - **Cardiac arrest** (ventricular fibrillation [VF], pulseless ventricular tachycardia [VT], asystole, pulseless electrical activity) (algorithm 2)
  - **Bradycardia with pulse** (algorithm 3)
  - **Tachycardia with pulse** (algorithm 4)
- **Team performance during resuscitation** – Teams providing ACLS perform better when there is a single designated leader who asks for and accepts helpful suggestions from members of the team, and when the team practices clear, closed-loop communication. (See 'Resuscitation team management' above.)
- **Initial interventions** – Begin excellent cardiopulmonary resuscitation (CPR) immediately for any patient with suspected cardiac arrest. Excellent chest compressions have few interruptions, are delivered at the correct rate and depth, and allow complete chest recoil (table 1). Secondary interventions include performing ventilations, administering oxygen, establishing vascular access, initiating appropriate monitoring (cardiac, oxygen saturation, waveform end-tidal carbon dioxide [$EtCO_2$]), and obtaining an electrocardiogram (ECG). (See 'Initial management and ECG interpretation' above.)

During initial life support of adults, high-quality chest compressions take priority over ventilation (circulation, airway, breathing [C-A-B]). When ventilating the patient in cardiac arrest, give 100 percent oxygen, use low respiratory rates (approximately six to eight breaths per minute), and avoid hyperventilation, which is harmful. Ventilation using a bag-valve-mask or supraglottic airway is preferred when possible. (See 'Airway management' above.)

- **ECG interpretation** – For the purposes of ACLS, ECG interpretation is guided by three questions:
  - Is the rhythm fast or slow?
  - Are the QRS complexes wide or narrow?
  - Is the rhythm regular or irregular?
- **Arrhythmia management** – The basic approach and important aspects of management for each arrhythmia covered by the ACLS Guidelines are discussed in the text and summarized in the accompanying algorithms. Patients with VF or VT are defibrillated as rapidly as possible. For patients with effective respiration and a palpable pulse, treatment is determined by the ventricular rate (tachycardiac or bradycardia) and clinical assessment of overall stability (see 'Management of specific arrhythmias' above and 'Medications used during CPR' above):
  - **Cardiac arrest** (VF, pulseless VT, asystole, pulseless electrical activity) (algorithm 2) (see 'Pulseless patient in sudden cardiac arrest' above)

    A single biphasic defibrillation is the treatment for VF or VT. CPR should be performed until the defibrillator is charged and resumed immediately after the shock is given without pausing to recheck a pulse.
  - **Bradycardia with pulse** (algorithm 3) (see 'Bradycardia' above)
  - **Tachycardia with pulse** (algorithm 4) (see 'Tachycardia' above)

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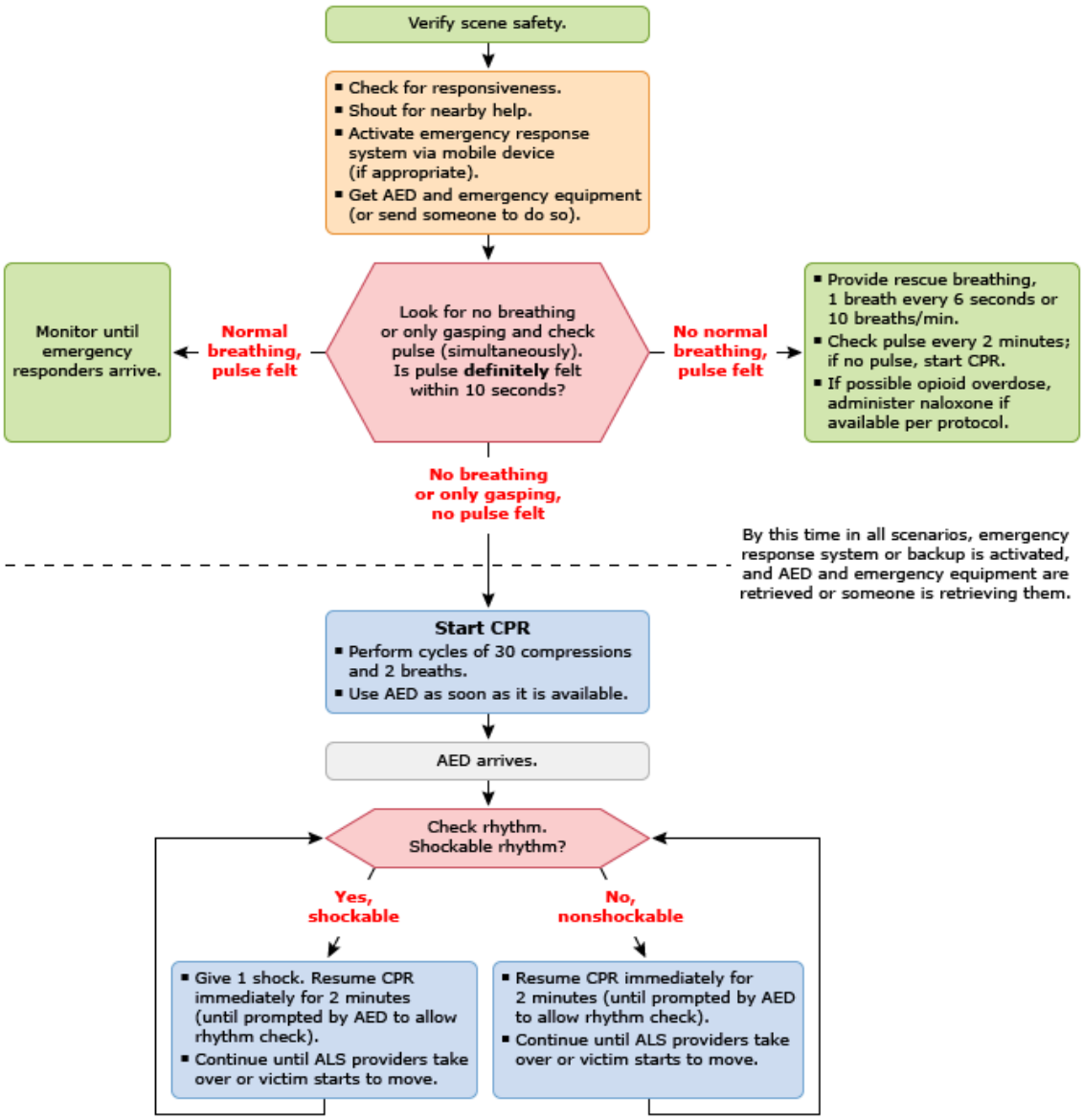
Topic 278 Version 108.0

GRAPHICS

Algorithm 1: **BLS health care provider adult cardiac arrest algorithm – 2020 update**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Adult Basic Life Support Algorithm for HealthCare Professionals is now available. This algorithm will be updated within UpToDate in the future.

AED: automated external defibrillator; ALS: advanced life support; BLS: basic life support; CPR: cardiopulmonary resuscitation.

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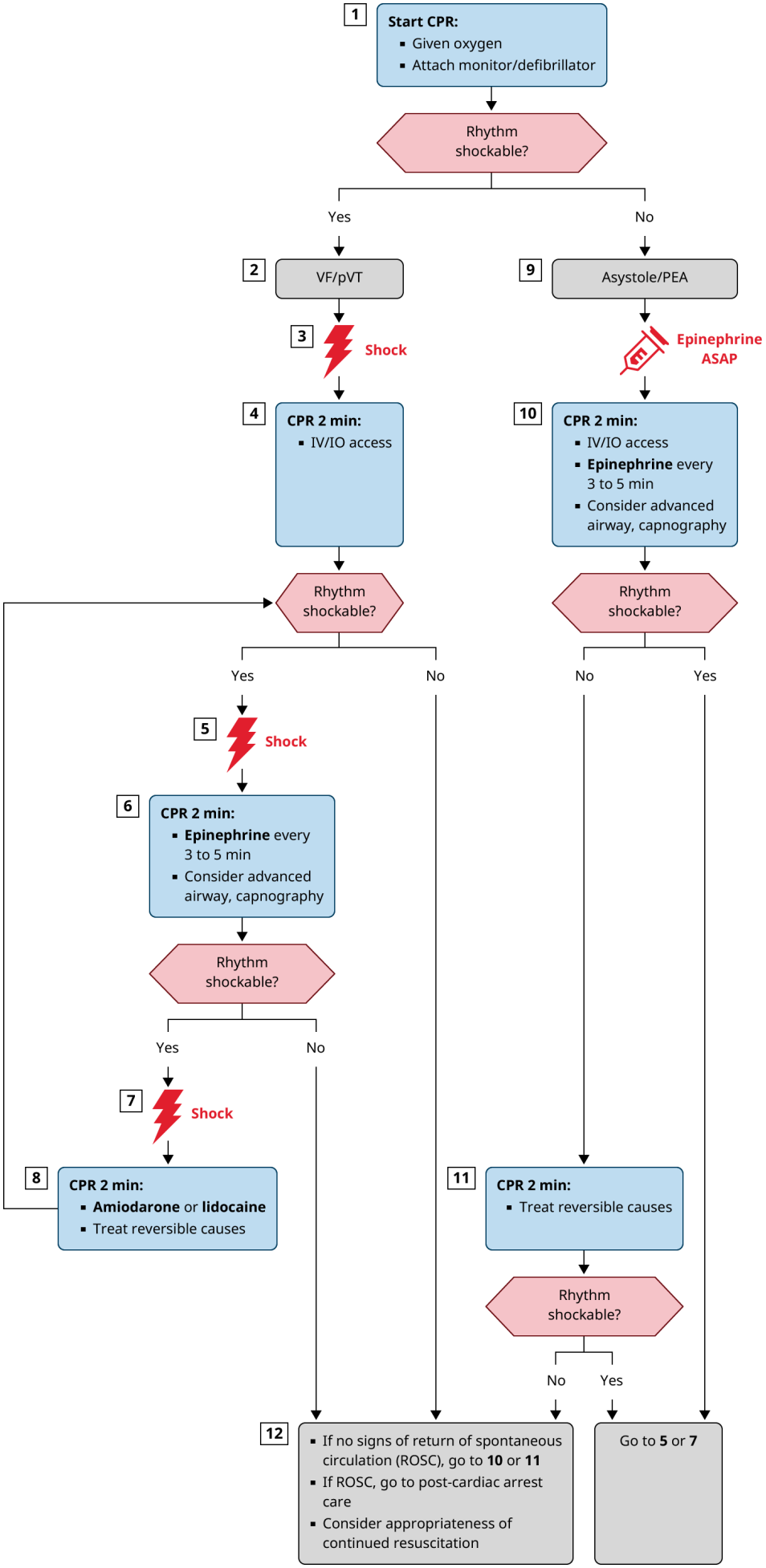
Graphic 105569 Version 13.0

Algorithm 2: **Adult cardiac arrest algorithm**



CPR quality

- Push hard (at least 2 inches [5 cm]) and fast (100 to 120/min) and allow complete chest recoil.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Change compressor every 2 minutes, or sooner if fatigued.
- If no advanced airway, 30:2 compression-ventilation ratio.
- Quantitative waveform capnography
  - If $PETCO_2$ is low or decreasing, reassess CPR quality.

Shock energy for defibrillation

- **Biphasic:** Manufacturer recommendation (eg, initial dose of 120 to 200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- **Monophasic:** 360 J.

Drug therapy

- **Epinephrine IV/IO dose:**
  1 mg every 3 to 5 minutes.
- **Amiodarone IV/IO dose:**
  First dose: 300 mg bolus.
  Second dose: 150 mg.

  or
- **Lidocaine IV/IO dose:**
  First dose: 1 to 1.5 mg/kg.
  Second dose: 0.5 to 0.75 mg/kg.

Advanced airway

- Endotracheal intubation or supraglottic advanced airway.
- Waveform capnography or capnometry to confirm and monitor ET tube placement.
- Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions.

Return of spontaneous circulation (ROSC)

- Pulse and blood pressure.
- Abrupt sustained increase in $PETCO_2$ (typically ≥40 mmHg).
- Spontaneous arterial pressure waves with intra-arterial monitoring.

Reversible causes

- **H**ypovolemia
- **H**ypoxia
- **H**ydrogen ion (acidosis)
- **H**ypo-/hyperkalemia
- **H**ypothermia
- **T**ension pneumothorax
- **T**amponade, cardiac
- **T**oxins
- **T**hrombosis, pulmonary
- **T**hrombosis, coronary

Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Adult Cardiac Arrest Algorithm is now available. This algorithm will be updated within UpToDate in the future.

ET: endotracheal; IO: intraosseous; IV: intravenous; J: joules; PEA: pulseless electrical activity; $PETCO_2$: peak end-tidal carbon dioxide; pVT: pulseless ventricular tachycardia; ROSC: return of spontaneous circulation; VF: ventricular fibrillation.

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Graphic 129983 Version 14.0

Table 1: **Key principles in the performance of ACLS**

Excellent CPR is crucial.
▪ Anything short of excellent CPR does not achieve adequate cerebral and coronary perfusion. ▪ Excellent chest compressions **must** be performed throughout the resuscitation without interruption, using proper timing (100 to 120 compressions per minute) and force (5 to 6 cm [2 to 2.5 inches] depth), and allowing for complete chest recoil. ▪ Excellent chest compressions take priority over ventilation. ▪ If a second rescuer is present, ventilations must be performed using proper timing (6 to 8 breaths per minute in the intubated patient; ratio of 30 compressions to 2 ventilations if not intubated) and force (deliver each breath over one second, and only until chest begins to rise). Avoid hyperventilation. ▪ Do **not** stop compressions until the defibrillator is fully charged.
Defibrillate VF and pulseless VT as rapidly as possible.
Rapidly identify and treat causes of non-shockable arrest (PEA, asystole).
▪ Important causes include the 5 H's and 5 T's: Hypoxia, Hypovolemia, Hydrogen ions (acidosis), Hyper/Hypo-kalemia, Hypothermia; Tension pneumothorax, Tamponade-cardiac, Toxins, Thrombosis-coronary (MI), Thrombosis-pulmonary (PE). ▪ If immediately reversible causes (eg, tension pneumothorax, cardiac tamponade) are not corrected rapidly, the patient has little chance of survival.

ACLS: advanced cardiac life support; CPR: cardiopulmonary resuscitation; VF: ventricular fibrillation; VT: ventricular tachycardia; PEA: pulseless electrical activity; MI: myocardial infarction; PE: pulmonary embolism.

Graphic 83671 Version 5.0

Table 2: **Manual defibrillation performance bundle**

1. Attach and charge the defibrillator while continuing excellent chest compressions. 2. Stop compressions and assess rhythm (should take no more than 5 seconds). 3. If VF or VT is present, deliver shock; if non-shockable rhythm is present, resume excellent CPR (and clear the charged defibrillator*). 4. Resume excellent chest compressions and CPR **immediately** after the shock is delivered.
Critical point: Interruptions in chest compressions must be kept to a minimum: Do **NOT** stop compressions while defibrillator is being charged.

*The defibrillator is charged during CPR in anticipation of treating a shockable arrhythmia and to minimize interruptions in CPR. If a non-shockable arrhythmia is present, the charged defibrillator should be discharged into the machine, rather than the defibrillation pads, according to the manufacturer's instructions.

VF: ventricular fibrillation; VT: pulseless ventricular tachycardia.

Graphic 83670 Version 7.0

Table 3: **Treatable conditions associated with cardiac arrest**

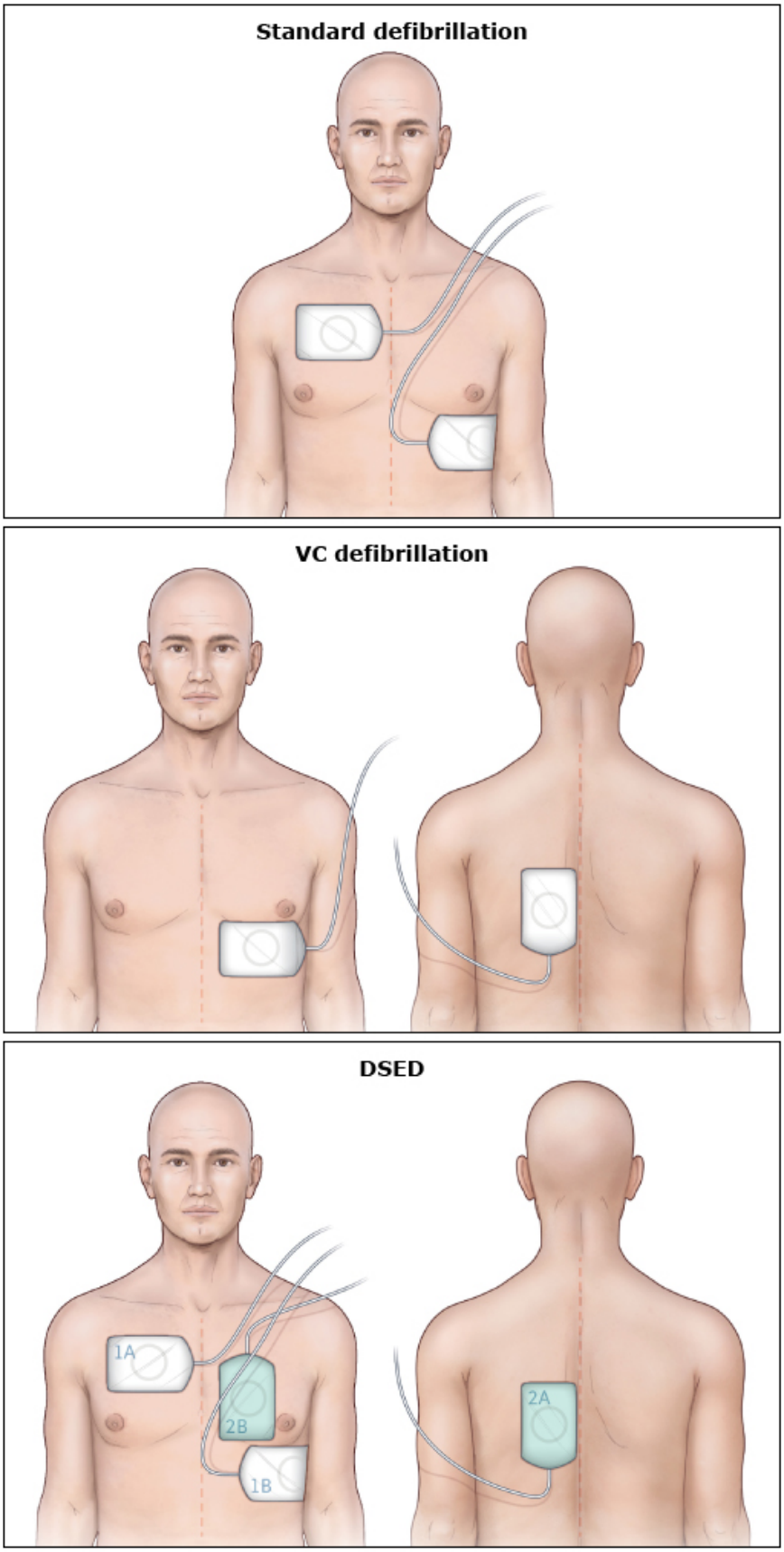
Condition	Common associated clinical settings
Acidosis	Diabetes, diarrhea, drug overdose, renal dysfunction, sepsis, shock
Anemia	Gastrointestinal bleeding, nutritional deficiencies, recent trauma
Cardiac tamponade	Post-cardiac surgery, malignancy, post-myocardial infarction, pericarditis, trauma
Hyperkalemia	Drug overdose, renal dysfunction, hemolysis, excessive potassium intake, rhabdomyolysis, major soft tissue injury, tumor lysis syndrome
Hypokalemia*	Alcohol abuse, diabetes mellitus, diuretics, drug overdose, profound gastrointestinal losses
Hypothermia	Alcohol intoxication, significant burns, drowning, drug overdose, elder patient, endocrine disease, environmental exposure, spinal cord disease, trauma
Hypovolemia	Significant burns, diabetes, gastrointestinal losses, hemorrhage, malignancy, sepsis, trauma
Hypoxia	Upper airway obstruction, hypoventilation (CNS dysfunction, neuromuscular disease), pulmonary disease
Myocardial infarction	Cardiac arrest
Poisoning	History of alcohol or drug abuse, altered mental status, classic toxidrome (eg, sympathomimetic), occupational exposure, psychiatric disease
Pulmonary embolism	Immobilized patient, recent surgical procedure (eg, orthopedic), peripartum, risk factors for thromboembolic disease, recent trauma, presentation consistent with acute pulmonary embolism
Tension pneumothorax	Central venous catheter, mechanical ventilation, pulmonary disease (eg, asthma, chronic obstructive pulmonary disease), thoracentesis, thoracic trauma

CNS: central nervous system.
* Hypomagnesemia should be assumed in the setting of hypokalemia, and both should be treated.

Adapted from: Eisenberg MS, Mengert TJ. Cardiac resuscitation. N Engl J Med 2001; 344:1304.

Graphic 52416 Version 8.0

Figure 1: **Pad placement for defibrillation**



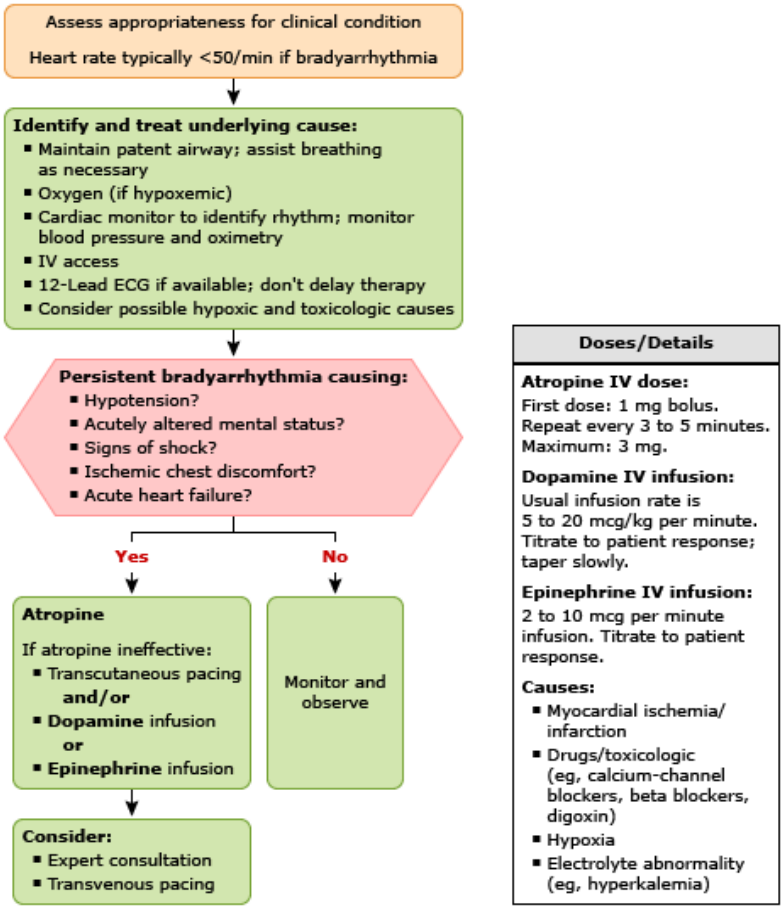
Vector change defibrillation consists of altering the direction of the vector. In clinical practice, this typically means moving the electrode pads from their standard position to an anterior-posterior position, as shown above. Double sequential external defibrillation (DSED) consists of rapid sequential shocks from 2 defibrillators.

DSED: double sequential external defibrillation; VC: vector change.

From: Cheskes S, Verbeek PR, Drennan IR, et al. Defibrillation strategies for refractory ventricular fibrillation. N Engl J Med 2022; 387:1947. Copyright © 2022 Massachusetts Medical Society. Reprinted with permission from Massachusetts Medical Society.

Graphic 103268 Version 3.0

Algorithm 3: **Adult bradycardia algorithm 2020 update**



Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Adult Bradycardia with a Pulse Algorithm is now available. This algorithm will be updated within UpToDate in the future.

ECG: electrocardiogram; IV: intravenous.

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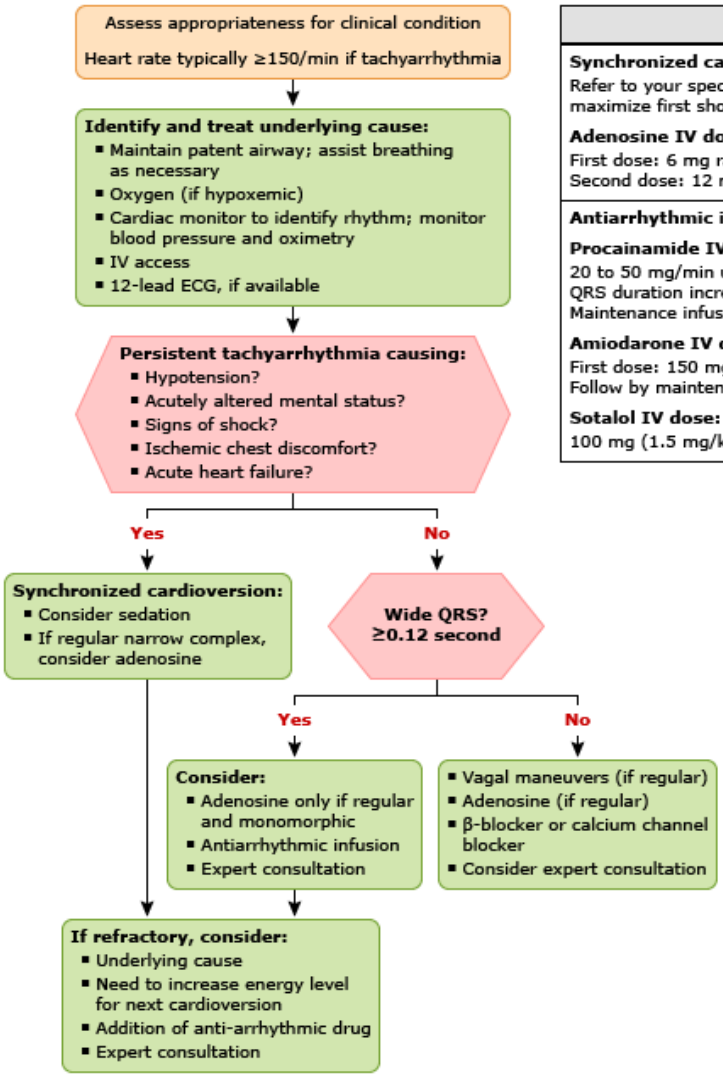
Graphic 130748 Version 16.0

Algorithm 4: **Adult tachycardia with a pulse algorithm 2020 update**



Doses/Details
Synchronized cardioversion: Refer to your specific device's recommended energy level to maximize first shock success. **Adenosine IV dose:** First dose: 6 mg rapid IV push; follow with NS flush. Second dose: 12 mg if required.
Antiarrhythmic infusions for stable wide-QRS tachycardia **Procainamide IV dose:** 20 to 50 mg/min until arrhythmia suppressed, hypotension ensues, QRS duration increases >50%, or maximum dose 17 mg/kg given. Maintenance infusion: 1 to 4 mg/min. Avoid if prolonged QT or CHF. **Amiodarone IV dose:** First dose: 150 mg over 10 minutes. Repeat as needed if VT recurs. Follow by maintenance infusion of 1 mg/min for first 6 hours. **Sotalol IV dose:** 100 mg (1.5 mg/kg) over 5 minutes. Avoid if prolonged QT.

Important notice: In October 2025, the American Heart Association released new guidelines on cardiopulmonary resuscitation and emergency cardiovascular care. An updated version of the Adult Tachyarrhythmia With a Pulse Algorithm is now available. This algorithm will be updated within UpToDate in the future.

CHF: congestive heart failure; ECG: electrocardiogram; IV: intravenous; VT: ventricular tachycardia.

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Graphic 130747 Version 16.0

Contributor Disclosures

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Conflict of interest policy

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